HDOC Day-50

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Problem Statement

Write a C program that reads the size and elements of a variable-length integer array **nums[]**. Pass the array to a function named **majority_element()** using its address and use pointers to pass the array between the calling function and the function.

1. Count the occurrences of each distinct element in nums[].

2. Find the element that appears strictly more than floor(size / 2) times.

3. Display the majority element if it exists.

4. Display -1 if no majority element exists.

Note: The majority element is not necessarily the element with the highest frequency. It must occur strictly more than floor(size / 2) times.

C Program

```c
#include <stdio.h>

void majority_element(int *nums, int size)
{
    int i, j, count;

    for (i = 0; i < size; i++)
    {
        count = 0;

        for (j = 0; j < size; j++)
        {
            if (*(nums + i) == *(nums + j))
            {
                count++;
            }
        }

        if (count > size / 2)
        {
            printf("Majority element: %d\n", *(nums + i));
            return;
        }
    }

    printf("-1\n");
}

int main()
{
    int size, i;

    printf("Enter the size of array: ");
    scanf("%d", &size);

    int nums[size];

    printf("Enter the array elements:\n");
    for (i = 0; i < size; i++)
    {
        scanf("%d", &nums[i]);
    }

    majority_element(nums, size);
    return 0;
}
```

Execution / Output

The following screenshot shows compilation and execution of the Day-50 program. For the input array **1 4 6**, no element occurs more than floor(3/2) = 1 time, so the program correctly displays **-1**.

